

$$\bar{i}_1 = \bar{i}_4 + \bar{i}_3 \rightarrow \frac{V_i - V_x}{R_1} = \frac{V_x}{R_4} + \frac{V_x}{R_3}$$

$$\left(R_3 + R_4 + \frac{R_3 R_4}{R_1} \right) V_x = \frac{R_3 R_4}{R_1} V_i$$

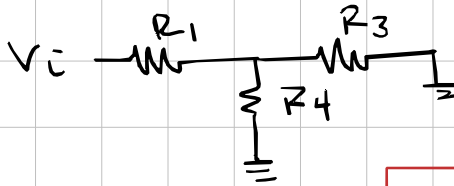
$$V_x = R_3 R_4 V_i / (R_1 R_3 + R_1 R_4 + R_3 R_4)$$

$$\bar{i}_3 = \bar{i}_2$$

$$\frac{V_x}{R_3} = \frac{0 - V_o}{R_2} \rightarrow \frac{R_4 V_i}{R_1 R_3 + R_1 R_4 + R_3 R_4} = -\frac{V_o}{R_2}$$

$$A_{ideal} = -R_2 R_4 / (R_1 R_3 + R_1 R_4 + R_3 R_4)$$

b) Ideal $R_i \rightarrow$ virtual short still exists:



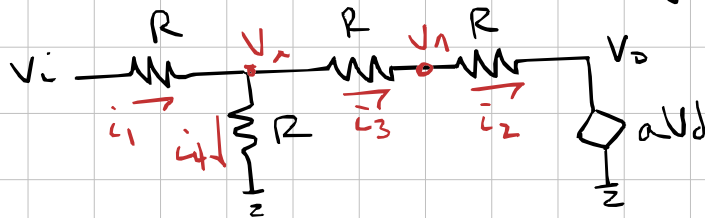
$$R_i = R_1 + R_3 \parallel R_4$$

$$R_i = R_1 + R_3 R_4 / (R_3 + R_4)$$

$$r_d = \infty, r_o \neq 0, a \neq \infty, R_1, R_2, R_3, R_4 = R$$

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c) $r_o = 0$, find A via exact analysis



$$V_o = aV_d \\ \rightarrow V_n = -V_o/a$$

$$i_1 = i_3 + i_4 \rightarrow \frac{V_i - V_x}{R} = \frac{V_x}{R} + \frac{V_x - V_o}{2R}$$

$$V_i - V_x = V_x + (V_x - V_o)/2$$

$$V_i = \frac{5}{2}V_x - \frac{V_o}{2}$$

$$V_n = -V_o/a = (V_x - V_o) \frac{R}{R+R} + V_o = (V_x - V_o)/2 + V_o$$

$$-V_o/a = \frac{V_x}{2} + \frac{V_o}{2}$$

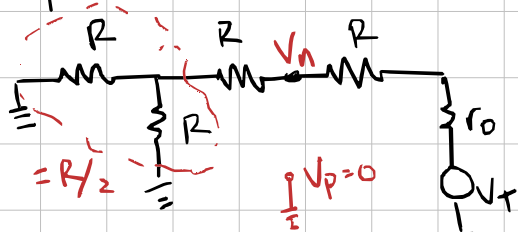
$$V_x = -(1 + 2/a)V_o$$

$$V_i = -\frac{5}{2}(1 + \frac{2}{a})V_o - \frac{V_o}{2} = -(\frac{6}{2} + \frac{5}{a})V_o$$

$$A = V_o/V_i = -\frac{1}{3} / (1 + 5/3a)$$

e) find T

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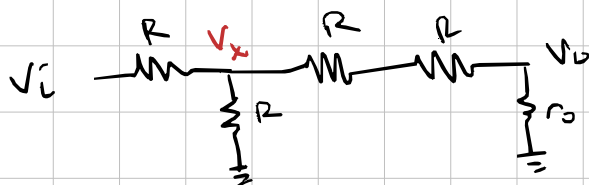


$$V_n = \frac{R + R/2}{R + R/2 + R + r_o} V_T = \frac{3/2}{5/2 + r_o/R} V_T$$

$$T = -a V_D / V_E = a V_n / V_T = a^{3/2} / (5/2 + r_o/R)$$

f) find a_{ft}

$$= a^{3/5 + 2r_o/R}$$



$$\begin{aligned} V_x &= R \parallel (R + R + r_o) / (R + R \parallel (R + R + r_o)) V_i \\ &= R(2R + r_o) / (R(3R + r_o) + R(2R + r_o)) V_i \\ &= (2R + r_o) / (5R + 2r_o) V_i \end{aligned}$$

$$V_o = V_x \cdot r_o / (2R + r_o) = r_o / (5R + 2r_o) V_i$$

$$a_{ft} = r_o / (5R + 2r_o)$$

$$g) A = A_{ideal} / (1 + 1/T) + a_{ft} / (1 + T)$$

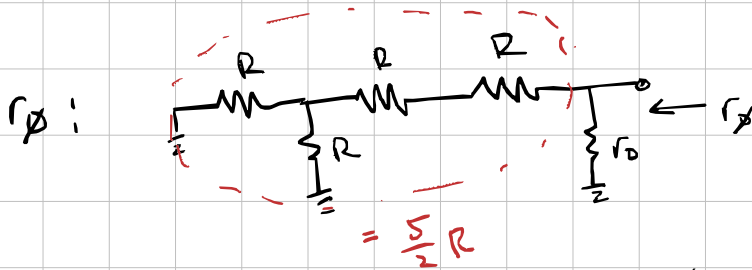
$$\begin{aligned} A_{ideal} &= -R_2 R_4 / (R_1 R_3 + R_1 R_4 + R_2 R_4) \\ A_{ideal} &= -1/3 \quad R_{1,2,3,4} = R \end{aligned}$$

$$A = -\frac{1}{3} / \left(1 + \frac{5/2 + r_o/R}{3/2 a}\right) + \frac{r_o / (5R + 2r_o)}{1 + a^{3/2} / (5/2 + r_o/R)}$$

validating with (c): set $r_o = 0$

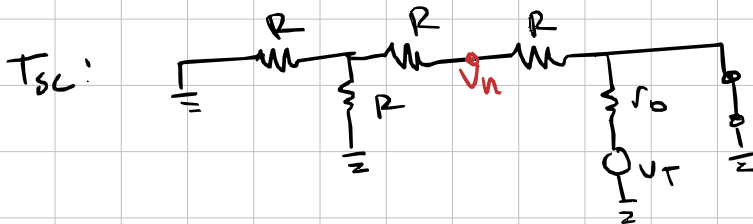
$$A = -\frac{1}{3} / \left(1 + \frac{5/2}{3/2 a}\right) + 0 = -\frac{1}{3} / \left(1 + \frac{5}{3a}\right)$$

h) find R_0 solution uses Blackman's formula, could do Directly...

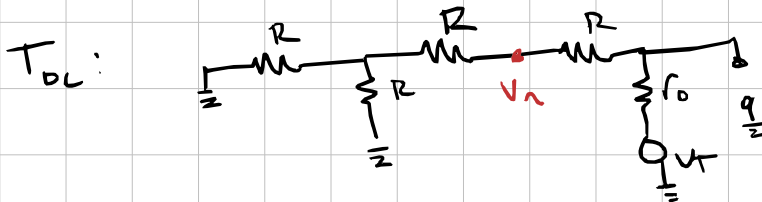


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$$r_\phi = \frac{5}{2} R \parallel r_o = \frac{\frac{5}{2} R r_o}{\frac{5}{2} R + r_o} = \frac{\frac{5}{2} r_o}{\frac{5}{2} + \frac{r_o}{R}}$$



$$v_n = 0 \rightarrow T_{SL} = 0$$



Same as in (e) $\rightarrow T_{DL} = \frac{\alpha^{3/2}}{\frac{5}{2} + \frac{r_o}{R}}$

$$R_0 = \frac{\frac{5}{2} r_o}{\frac{5}{2} + \frac{r_o}{R}} \left[\frac{1 + 0}{1 + \frac{\alpha^{3/2}}{\frac{5}{2} + \frac{r_o}{R}}} \right]$$

$$R_0 = \frac{\frac{5}{2} r_o}{\frac{5}{2} + \frac{r_o}{R} + \frac{\alpha^3}{2}}$$

i) find a_2 . Note: a_2 implies $r_0 = 0$ otherwise we would have T!

→ want $L = a_2 b$

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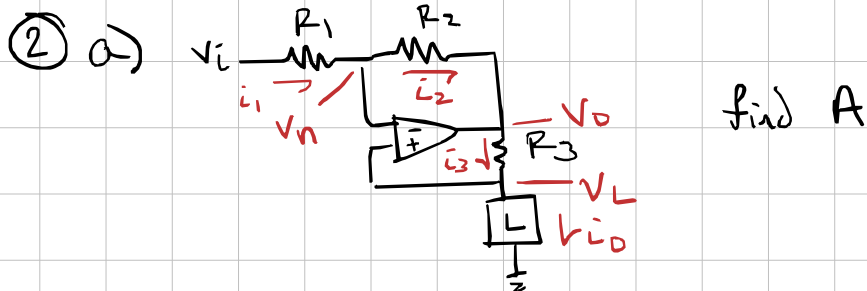
from c) or g) with $r_0 = 0$

$$A = -\frac{1}{3} / 1 + 5/3a \quad \therefore L = 3a/5$$

$$A_{ideal} = -\frac{1}{3} \rightarrow b = -3$$

$$L = a_2 b \Rightarrow \frac{3a}{5} = a_2 (-3)$$

$$a_2 = -a/5$$



By virtual short: $V_n = V_L$

$$\bar{i}_3 = \bar{i}_o = \frac{V_o - V_L}{R_3} \quad \bar{i}_1 = \bar{i}_2 \Rightarrow \frac{V_i - V_L}{R_1} = \frac{V_L - V_o}{R_2}$$

$$\downarrow \quad V_o - V_L = \frac{R_2}{R_1} (V_L - V_i)$$

$$\bar{i}_o = \frac{R_2}{R_1 R_3} (V_L - V_i)$$

$$A = -\frac{R_2}{R_1 R_3} \quad R_o = -\frac{R_1 R_3}{R_2}$$

b) $V_L = 2V$, $\bar{i}_o = 4mA$, $R_1, R_2, R_3 = 1k\Omega$, $\bar{i}_3 = 2mA$, $V_S = \pm 10V$

find V_o & $\bar{i}$ out of amplifier $\bar{i}_o'$

$$\bar{i}_o = \frac{V_o - V_L}{R_3} \rightarrow V_o = (4mA)(1k) + V_L = 6V$$

$$\bar{i}_o' = -\bar{i}_2 + \bar{i}_3 = \frac{V_o - V_L}{R_2} + \frac{V_o - V_L}{R_3} = 2\bar{i}_o = 8mA$$

$$P_{out} = (10V - 6V)(8mA) = 32mW$$

$$P_g = (10V - (-10V))(2mA) = 40mW$$

$$P = 72mW$$

③ $A = 100 \text{ V/V}$, $\text{CMRR} = 80 \text{ dB}$

$$V_1 = 2.47 \text{ V}$$

$$V_2 = 2.56 \text{ V}$$

Since CMRR fairly large, know: $A_{DM} \approx A$
and $A_{CM} \ll A_{DM}$

$$\text{CMRR} = 80 \text{ dB} = 10000 = \frac{A_{DM}}{A_{CM}} = \frac{100}{A_{CM}}$$
$$A_{CM} = 0.01$$

$$V_{out} = A_{DM} V_{DM} + \underbrace{A_{CM} V_{CM}}_{\text{error}}$$

$$A_{CM} V_{CM} = 0.01 (2.56 + 2.47) / 2$$

$$\text{error} = 25.2 \text{ mV} \rightarrow 2.8\%$$

for reference, $i_{\text{ref}} = 100 (2.56 - 2.47) = 9 \text{ V}$

Actual output: 9.0252 V